HDOC Day-68: Decimal to Binary Using Recursion in C

1. Aim

To write a C program to convert a decimal number into its binary representation using a recursive function.

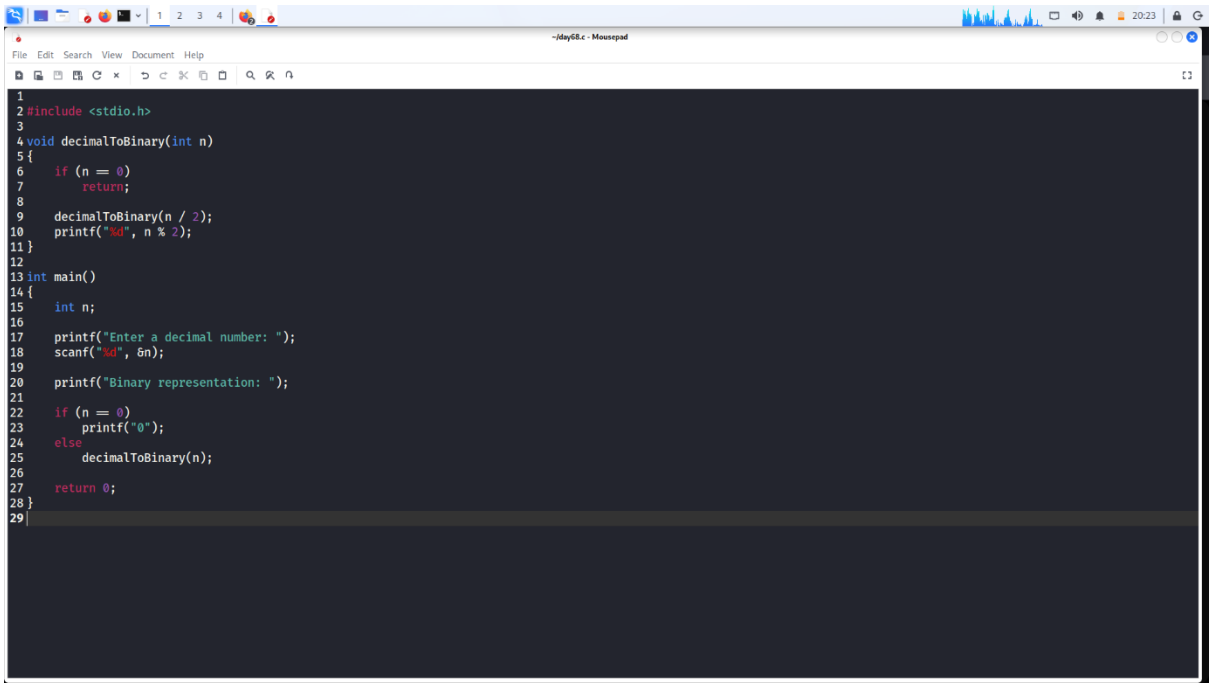
2. Question

Write a recursive function to convert a decimal number into its binary representation.

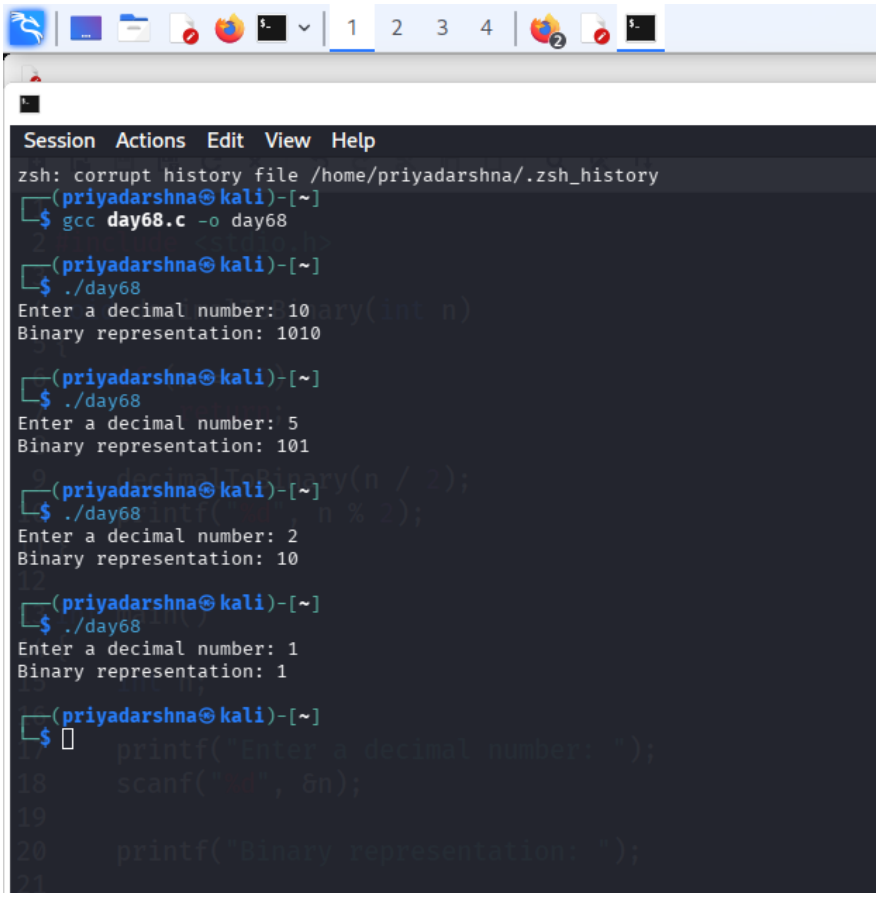
3. Algorithm

1. Start the program.
2. Declare an integer variable n.
3. Read the decimal number from the user.
4. Define a recursive function decimalToBinary(n).
5. If n == 0, return from the function.
6. Otherwise, call decimalToBinary(n / 2).
7. Print the remainder n % 2.
8. Stop the program.

>C Program:



>OUTPUT:


1 2 3 4
Session Actions Edit View Help
zsh: corrupt history file /home/priyadarshna/.zsh_history
(priyadarshna㉿kali)-[~]
$ gcc day68.c -o day68
(priyadarshna㉿kali)-[~]
$ ./day68
Enter a decimal number: 10
Binary representation: 1010
(priyadarshna㉿kali)-[~]
$ ./day68
Enter a decimal number: 5
Binary representation: 101
(priyadarshna㉿kali)-[~]
$ ./day68
Enter a decimal number: 2
Binary representation: 10
(priyadarshna㉿kali)-[~]
$ ./day68
Enter a decimal number: 1
Binary representation: 1
(priyadarshna㉿kali)-[~]
$